MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5112/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the lithium sulfur dioxide non-rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49430(ER).

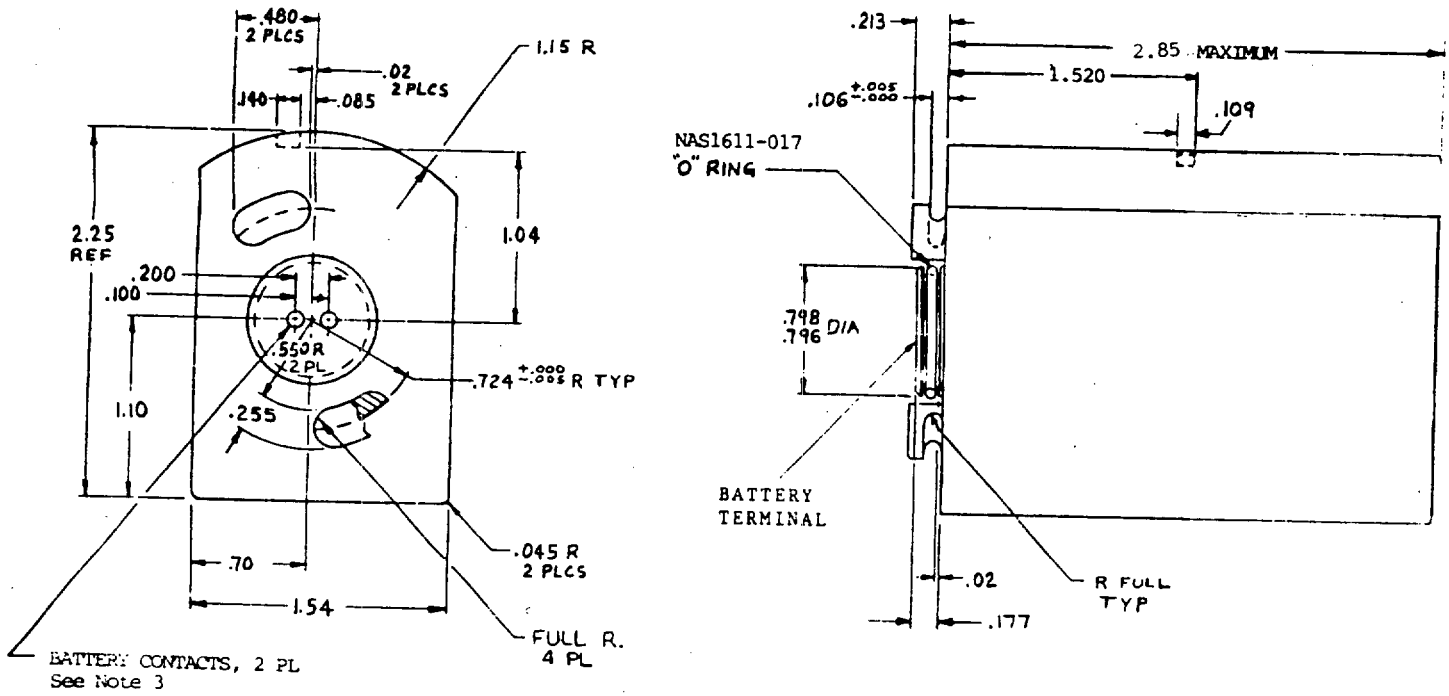


Figure 1. Battery dimensions.

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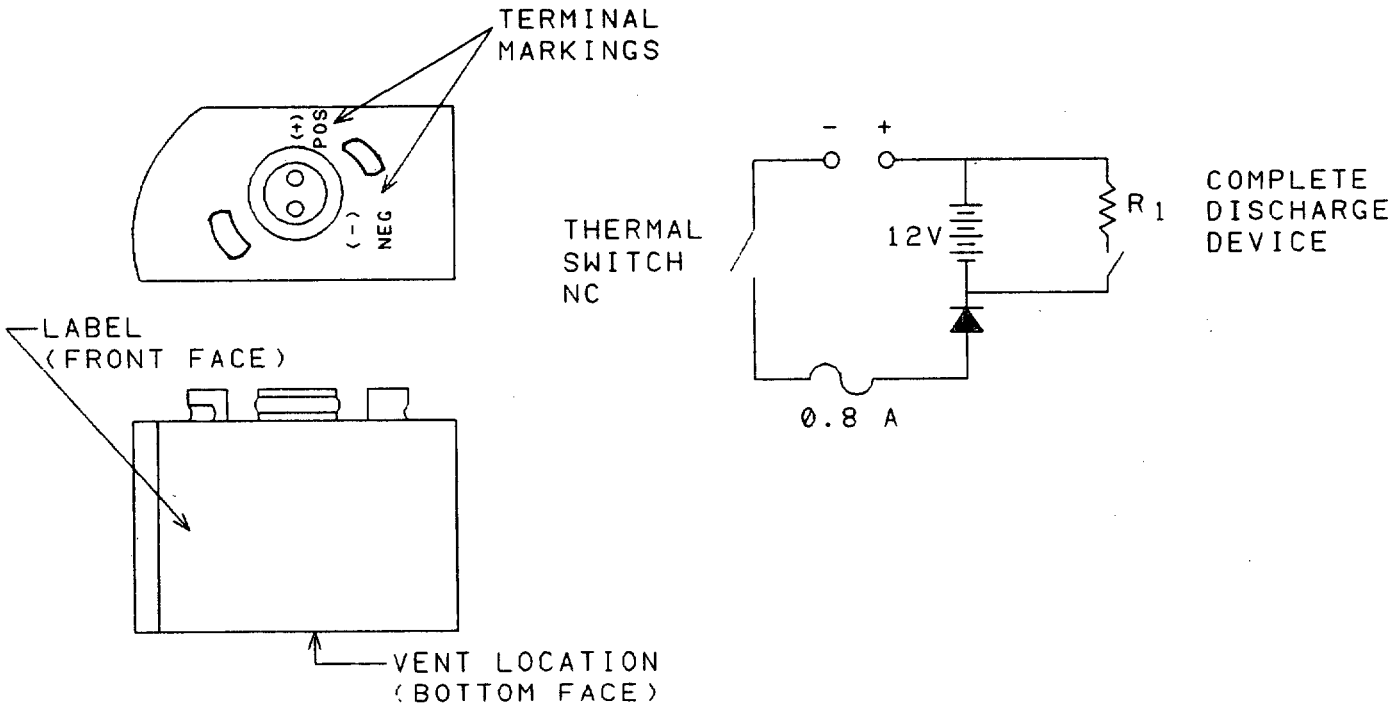


FIGURE 2

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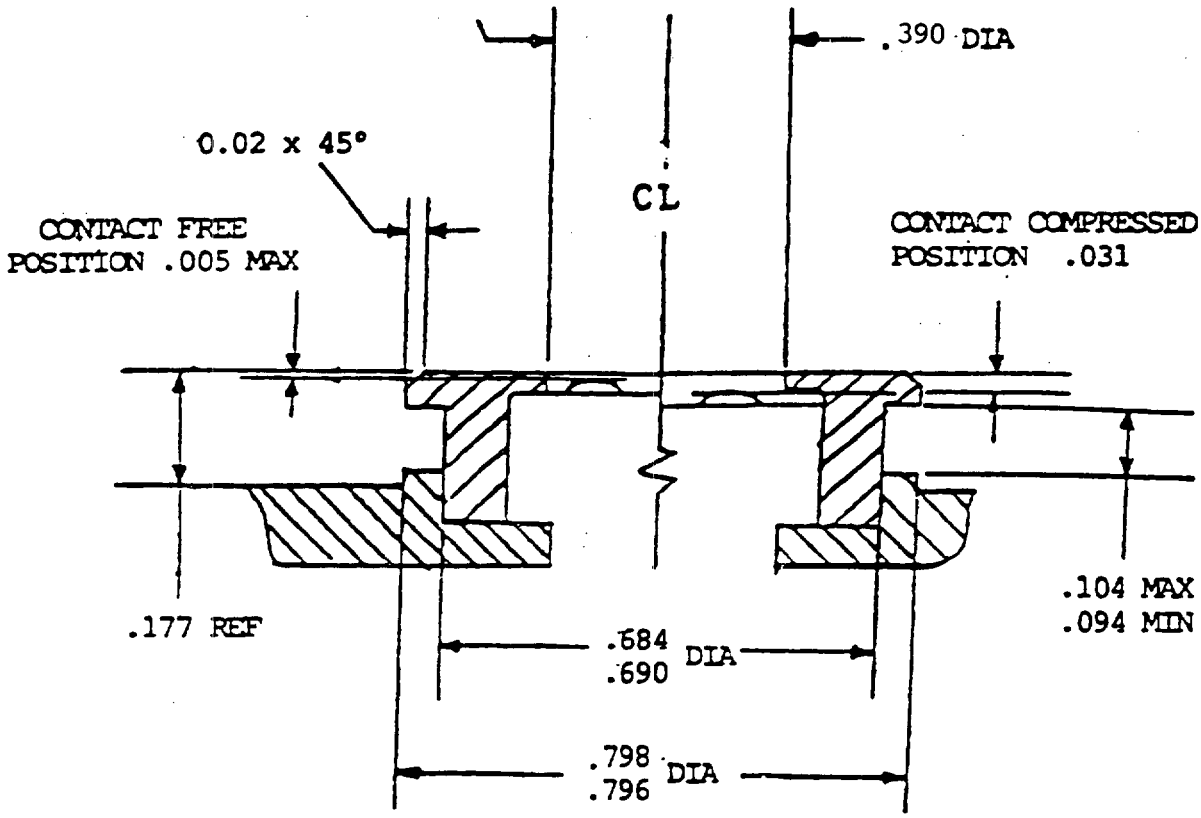


FIGURE 3

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NOTES:

1. All dimensions are in inches.

2. Tolerances are: 2 place decimals ±.01.
3 place decimals ±.005.

3. Battery contacts shall be constructed from tubular, oval head rivets conforming to MS16535-11. The contacts shall be nickel plated in accordance with QQ-N-290, Class 1, Grade C. Then the contacts shall be gold plated in accordance with MIL-G-45204, Type II, Grade C, Class 2 approximately .00144 grams, .0456 square inches. The contacts shall be mounted on a spring loaded surface. The contacts shall require a minimum mating force of 15 ounces per contact. The contacts shall not protrude above the surface of the battery terminal. The contacts diameter shall be .109\.005 inch and the contact surface shall be smooth.

4. R_1 value shall be between 370 ohms and 520 ohms and have a power rating of 0.5 watt.

5. Connector wiring schematic to be printed or embossed on battery case in suitable location as indicated.

REQUIREMENTS:

Dimensions and configuration: See Figure 1 and Figure 3.

Maximum voltage: 12.20 volts.

Minimum voltage for capacity: 8.00 volts.

Electrical Fuse: The battery shall contain an 0.80 ampere slow blow fuse non replaceable in series with the cells in the negative leg.

Diode: The battery shall contain a blocking diode wired in series with the cells. See paragraph 3.5.5.2 of the Basic Specification.

High Temperature Switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in paragraphs 3.5.5 and 3.5.5.1 of the Basic Specification.

Terminals: 2 pin contact, coupler interface, "O" ring sealed (see Figure 1 and 3).

Weight (maximum): 0.40 pounds (180 gms).

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 8.00 Volts
I	3.00
L	1.50
H	5.50
HT	5.30
LT	1.30
T	2.80
D	2.80
ILR	1000

Initial voltage delay: During the I, H, HT, T, D, and ILR tests, initial voltages below the 8.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 8.90 volts cannot exceed a 60.0 second duration .

Venting Mechanism: The battery case shall incorporate a venting mechanism that shall prevent the battery case from expolding or rupturing into fragments when cell venting occurs. The venting mechanism shall be located at the bottom face (see Figure 2.0). Venting mechanism shall not become separated from the battery case when activated by cell venting.

O-Ring seal: The battery when subjected to the O-Ring seal test shall not increase in weight by more than 0.1 grams and the O-Ring shall prevent the water from contacting the battery contacts and the battery terminal face. There shall be no sign of corrosion on the battery contacts.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para. 3.5.5.3 of the basic specification.

METHOD OF EXAMINATION AND TEST:

Capacity tests:

(1.) Storage: Details on storage conditions for all specified capacity tests are described in the basic specification.(see 4.8.9)

(2.) Discharge:

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Cells: Discharge at 0.65 amps constant current at 70±5°F for a minimum of 2.7 hours to 2.00 volts cut-off. Continue discharge to zero volts, then discharge at 0.65 amperes constant current for 2.7 hours into voltage reversal.

Batteries: For all capacity tests except the H, HT, and ILR, the batteries shall be discharged at 0.65 amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through 0.65 amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I test the batteries shall be forced discharge at 0.65 amperes constant current for 3.0 hours immediately after zero volts is obtained. There shall be no venting, fire or explosion when the batteries are forced discharged. For the ILR test a fixed resistance of 5600 ohms shall be used to a cut-off voltage of 8.00 volts or 1000 hours.

Closed circuit voltage: Load resistance of 20 ohms shall be used between battery terminals. Voltage shall be above 8.0 volts within 5 seconds.

Venting Mechanism: The battery case venting mechanism shall be monitored during the Safety Feature Test. During cell venting the battery venting mechanism shall release the pressure build up in the battery case and shall meet the performance requirements specified herein.

O-ring seal test: Five samples shall be selected from the First Article Group I samples. These samples shall be selected prior to the capacity tests. The weight of each battery sample shall be measured and recorded. The samples shall be mated to a Motorola part number 64-P21451J001 coupler plate or equal. The coupler plate shall be waterproofed, except at the battery/coupler plate interface, to prevent water from contacting the battery connector contacts. The battery/plate assemblies shall be immersed in a synthetic solution containing 3.6 percent by weight sea salt in accordance with ASTM-D-1141. The solution with the battery samples shall be pressurized to simulate two foot immersion for 24 hours and the pressure shall be raised to simulate 50 foot immersion for 15 minutes. The battery samples shall be removed from the solution and separated from the coupler plate. The battery contacts shall be inspected for evidence of water. The batteries shall be towel dried and left to air dry for 15 minutes. The batteries shall be weighed and shall meet the O-ring seal requirements specified.

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